A REAL-TIME MOVEMENT SONIFICATION APPLICATION FOR BODYWEIGHT SQUAT TRAINING

Cherelle Connor, Prithvi Kantan, Stefania Serafin

Aalborg University
Copenhagen, Denmark
{cco,prka,sts}@create.aau.dk

ABSTRACT

The bodyweight squat is an accessible, versatile and effective movement, fundamental to common activities of daily living such as: standing, sitting, and lifting. Yet, many individuals experience difficulty understanding how to perform this movement correctly. Movement sonification is a reliable method for providing guidance during physical activity. We developed an auditory-only exercise application that provides guidance for squat training through the use of real-time movement sonification feedback. We provided feedback for four movement parameters of the squat: foot placement, knee flexion angle, knee alignment, and weight shifting. Two evaluations were conducted. In the first evaluation, four participants provided feedback on five sonic designs and mappings. We found that participants preferred contextual sonic designs opposed to continuous designs. In the second evaluation, four participants assessed the usability of the application. The application was found to have a mean System Usability Score of 78.75. Our work outlines the design of a real-time concurrent sonification scheme that provides logical, sequential feedback information for multiple movement parameters. We anticipate that this paradigm will assist in further establishing the potential of movement sonification to augment squat training when no visual input is present.

1. INTRODUCTION

Bodyweight training has been shown to improve strength, power, and endurance across many demographics [1]. Bodyweight exercises are not only beneficial for maintaining a healthy body, but also serve to strengthen and prepare our bodies for activities of daily living such as, sitting, standing, and lifting [2]. An inability to perform these movements can be an initial indicator of larger issues like misalignment or impingement of joints and muscles [3, 4]. Bodyweight exercises offer users a cost-effective, portable and space-saving way to exercise outside of a gymnasium [1]. In particular, the bodyweight half squat (henceforth referred to as *squat*) is an integral part of many athletic movements, and is a key exercise for strengthening lower-body muscles [5]. Bodyweight squats can strengthen several muscle groups at once, making it a common movement within clinical settings for lower body rehabilitation as well [1, 6].

Although the squat is a fundamental movement for activities of daily living [7, 8], many individuals lack knowledge on correct squat form, which can lead to suboptimal movement execution in the absence of a coach or trainer [9]. Recent advancements in sensing technologies have used interactive paradigms designed to encourage physical activity and fitness through automated trainers and guidance systems reducing the need for constant monitoring [10, 11].

Various data capture techniques, such as *inertial measurement units (IMUs)* and optical tracking devices, have been used to capture motion data and provide real-time feedback on task-relevant movement descriptors across haptic, visual, and auditory modalities [12, 13, 14]. The use of auditory feedback has several advantages over its visual counterpart. Visual feedback has the potential to overload visual perception and cognitive processing [12, 15]. Using auditory feedback techniques, in place of visual feedback, can potentially reduce cognitive workload and distraction, decreasing the amount of focused attention required from the user comparatively and offer a greater level of portability and cost effectiveness by reducing the need for visual displays [12, 15].

Auditory feedback generated by sonifying movement parameters in real-time can serve as an effective technique for improvement of physical rehabilitation, dance and sports technique performance. [16, 17, 18, 19] Movement sonification refers to the transformation of these movement parameters into sound. [20] The movement parameters are mapped to sound synthesis or processing parameters, creating an extrinsic feedback channel that can complement proprioceptive information, allowing for a greater understanding of how and where their body is moving in space [21].

1.1. Previous Approaches to Squat Correction

Sonification approaches providing real-time feedback on knee flexion have shown promise in few experimental studies. Newbold [22] devised a musically informed sonification scheme to support the squat movement, and aid in motivating the user to complete the exercise. They tracked the angle of a participant's leg via a smartphone strapped to their upper thigh, as they completed a set of squats. They assessed four sonification feedback conditions: stable and unstable musical cadences, non-musical feedback (noise) or no sound at all. Comparing these conditions, the authors found that participants could be motivated to squat beyond their target flexion point based on their expectancy of the musical cadences. They also saw reduction of movement beyond the target flexion point in the noise and no sound conditions. The use of noise as feedback in this study demonstrated how users can be quickly discouraged from performing undesirable movement patterns [22].

Hale et al. [23] designed and validated a squat training program that provided two movement parameter error sonification feedback. The movement parameters monitored were target knee angle flexion and center of pressure. Participants were split into trained (received auditory feedback) and control (no auditory feedback) groups during the 5-week study. All participants were fitted

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with an electric goniometer that measured knee flexion and a bilateral force plate that measured center of pressure. The feedback was provided via a custom designed LabVIEW program. The trained group received feedback that consisted of both concurrent feedback (during movement) and terminal feedback (after completion). The concurrent feedback was administered in the form of beeping sounds that indicated the precision of the target knee flexion angle and the center of pressure target location. The terminal feedback was administered in the form of either a cheering or booing sound dependent on if the participants performed the movement correctly. The control group did not receive any feedback. The trained group showed significant improvement in reaching their target knee flexion angle and maintaining ideal center of pressure location. This study demonstrated that auditory feedback is an effective way to train two targets simultaneously [24].

Due to the limited research for squat correction involving auditory feedback, we also reviewed some past visual feedback approaches involving other biomechanical aspects of the squat. Conner and Poor [10] created a graphical user interface designed for users to view their free weight squat corrections. Using the Kinect V2, the interface tracked the angle of the knee and position of the feet and shoulders throughout the squat movement. The interface provided terminal feedback on five requirements for ideal squat form: thighs parallel to the ground, flat back, flat feet, even weight distribution, and weight centered over ankles. Within the interface, on the right side of the screen, users were presented with a video feed of themselves performing the squat in real-time. On the left side of the screen was a list consisting of the five requirements used to track the performance of the squat. The interface indicated whether the listed requirements were fulfilled for each repetition after its completion. Exit interviews conducted with the participants of this study, found the software to be helpful and believed the feedback provided was useful in improving their squat form [10].

Sanford et. al. [25] investigated how complex representative visual feedback could improve movement consistency for a designated target trajectory during squat performance. In this study, complex representative visual feedback referred to feedback provided for three movement features: the torso, thigh and shank orientations. These orientations were tracked via the Optitrack motion capture system and the angle of the thigh during knee flexion was calculated. Muscle activation data was also collected using wireless EMG sensors. The feedback was displayed concurrently on a monitor in front of the participants, as a lower body skeletal illustration with the appropriate connections at joint locations. Participants evaluated the concurrent feedback in a training session, followed by a post-assessment session with no feedback. The feedback was shown to not only reduce the thigh angle error compared to its target trajectory during squat repetitions, but also to normalized the EMG magnitudes across the muscles involved in the training [25].

1.2. Problem Formulation

Based on these previous studies, we targeted several areas of squat form improvement that we believed would provide more immediate and accessible feedback for beginner recreational athletes, the elderly, and individuals with visual disabilities. Similar to Conner and Poor[10], our aims were to provide feedback on four squat-relevant variables: knee flexion, foot placement, knee alignment, and weight shifting. We chose these requirements as they are consistently problematic areas in squat training [7, 8, 26]. We provided augmented audio feedback for not one, but all four requirements identified. Currently, there are no other auditory feedback studies that have accomplished this.

Hale's study [23], showed that concurrent (simultaneous) and terminal auditory feedback are an effective means by which to train different target outcomes. We intend to use these findings to expand on this work and create a concurrent, sequential interaction that provides real-time, instantaneous task-relevant feedback during the descent and ascent stages of the squat. Although terminal feedback has been shown to be successful in improving retention rates of physical skills [23], this feedback does not address squat form errors until the movement has been completed. It is vital that user errors are addressed immediately. Incorrect squat form can cause undue stress on the ligaments and joints of the lower body and increase the chances of injuries or tears [7]. Therefore, it is crucial that concurrent feedback is used to notify users of their errors immediately, so they are not propagated through the remaining squat repetitions. Such a targeted and context-dependent sequential sonification strategy for training has not yet been thoroughly explored. This approach creates a well-paced, structured method of training and will enable the user to focus on different parameters of the movement that are relevant during the descent and ascent of the squat.

These steps will address some of our perceived shortcomings of past work and make squat training via sonification accessible to a wider range of users. Improvement in the listed areas will benefit our efforts to develop an auditory-only system that provides real-time movement sonification feedback for squat training that ensures the user is performing the exercise safely and correctly.

In the following sections, we report on the iterative process used to design this application. Once our initial prototype was developed, we conducted a brief evaluation to determine which sonic mappings were most preferable. With the results from the initial evaluation and the verbal feedback provided by the participants, we redesigned the prototype to include the full set of movement features in a scripted sequential feedback interaction. The second evaluation assessed the usability of the application.

2. ITERATION 1

2.1. Design

We first focused on designing and assessing mapping schemes to represent knee flexion during the squat, mainly due to its importance in acquiring the full benefits of the movement [8] and its relative ease of computation from optical tracking data.

2.1.1. Technical Development

The user's body motion was tracked using the Microsoft Azure Kinect Camera [1] connected to a PC via USB-C. Spatial information was extracted using the Kinect's Body Tracking SDK [2] at a frame-rate of 30 fps. Specifically, the SDK provided tracking information for 32 joints in the X, Y, and Z planes. The application interface was developed in Unity3D Game Engine [3], and sound

[1] https://docs.microsoft.com/en-us/azure/kinect-dk/

[2] https://docs.microsoft.com/en-us/azure/kinect-dk/body-joints

[3] https://unity.com/

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generation was carried out in the Faust[4] programming language (refer Fig. 1).

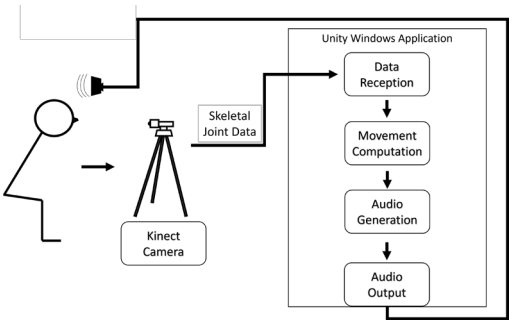


Figure 1: Signal flow diagram for the squat trainer application.

2.2. Implementation

2.2.1. Movement Feature Computation

The knee angle was calculated using the Law of Cosines,

$$b^2 = a^2 + c^2 - 2accos(B) \tag{1}$$

transformed to solve for angle B:

$$B = cos^{-1}((a^2 + c^2 - b^2)/2ac) \tag{2}$$

where a is the length on the thigh, b is the length between the hip and ankle joints, c is the length on the shank, and B is the calculated knee angle. The angle of the knee was reported in degrees. As the knee was extended, the angle increased, as the knee flexed, the angle decreased. Full knee extension corresponded to 180 degrees and full knee flexion corresponded to 90 degrees or greater depending on the user's range of motion.

2.2.2. Sonic Interaction Design and Mapping

We used five different sonic designs to map knee flexion. Two of the designs provided constant feedback, meaning the auditory feedback was present even as the user stopped moving or reached their target position. These designs were the Whistling Bottles and Non-linear Filter Modulated Sine Wave. The remaining three designs provided contextual feedback, meaning the auditory feedback reflected the context of the squat movement parameter it represented. These designs were the Conch Trumpet, Wooden Keyboard and Tuned Bars. All designs were based on physical models created in the Faust [4] programming language.

Whistling Bottles: This model consisted of a whistle instrument, designed to mimic the sound of blowing into a jug. The model contained parameters for whistle echo and reverberation on both the whistle and the "virtual room". The user's knee flexion modulated the reverberation volume of the whistle. As the participant approached their target knee flexion angle the reverberation volume decreased, making the sound of the whistle appear close to the user. As the knee was extended, the reverberation volume increased making the whistle sound further away. This mapping was intended to match the proximity of the user's target angle to the proximity of the sound. If the current angle was close to the target angle, the whistle sounded closer to the user.

Modulated Sine Wave: This model consisted of a sine wave. The model contained parameters for frequency ranging from 100HZ to 1200Hz. The user's knee flexion modulated the frequency of the sine wave. As the participant approached their target knee flexion angle the frequency of the sine wave increased. The frequency decreased as they extended their knee. This increase in pitch was intended to mimic the tension of the leg muscles as they flexed in the squat position.

Conch Trumpet: This model was based on a preexisting trumpet physical model. The model contained a parameter for blowing pressure, ranging from 0-1. The user's knee flexion modulated the blowing pressure of the air flow created by the virtual performer. As the participant approached their target knee flexion angle, the blowing pressure increased. The lower the the participant squatted the more intensely the horn would blow. The blowing pressure decreased to zero as the user returned to full knee extension. This mapping was intended to give the user a sense of control over the intensity of the sound produced during the squat movement.

Wooden Keyboard: This model was designed to mimic the notes produced when a mallet strikes multiple keys of a wooden keyboard consecutively. The model contained a parameter for "instrument hand" ranging from 1-10, which represented 10 keys of the keyboard. The sound produced is similar to a xylophone or marimba. The user's knee flexion modulated the striking of the keyboard. As the participant approached their target knee flexion angle the amount of strikes to the keyboard decreased to zero, at which, no sound was played. As their knee angle extended, the strikes to the keyboard increased. This mapping was intended to emphasize feelings of pulling the body down while descending into the squat position and pushing the body up while ascending into the standing position.

Tuned Bars: This model consisted of two cascading notes, reminiscent of a chime. The model contained a parameter for "speed", which represented the amount of time between notes. The user's knee flexion modulated the time between the two notes. As the participant approached their target knee flexion angle the time between the notes increased, resulting in the perception of the chime being played slower until it stopped entirely. As their knee angle extended, the the time between notes decreased, resulting in the perception of the chime being played quicker. This mapping was intended to give the user a sense of pace, calm, and pleasantness as they entered the squat position. Sample audio clips of these sonic designs can be found here[5].

2.3. Sonic Evaluation

The first evaluation served to compare the five aforementioned sonic mappings in terms of informativeness and user preferences.

2.3.1. Participants

Four male participants provided feedback on the sonic design. All of the participants had a background in sound and music computing. All participants were volunteers and consented to being included in the evaluation. No sensitive information was collected from any participant.

[4] https://faustide.grame.fr/

[5] https://drive.google.com/drive/folders/12yF28XDxeob0QeND0nhGaL0ZaAdcJ9Wj?usp=sharing

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2.3.2. Experimental Setup

The evaluation was carried out in a naturally lit room. Participants were requested to stand on a marker five feet away from the camera, with their body rotated at an 45 degree angle, so that their left leg was closest to the camera but their right leg was still visible. This stance resulted in the user's body being rotated from a sagittal view to a 3/4 view when viewed from the camera (refer Fig. 2). This position reduced the risk of occlusion by ensuring that both of the user's legs were visible by the camera, during the movement execution.

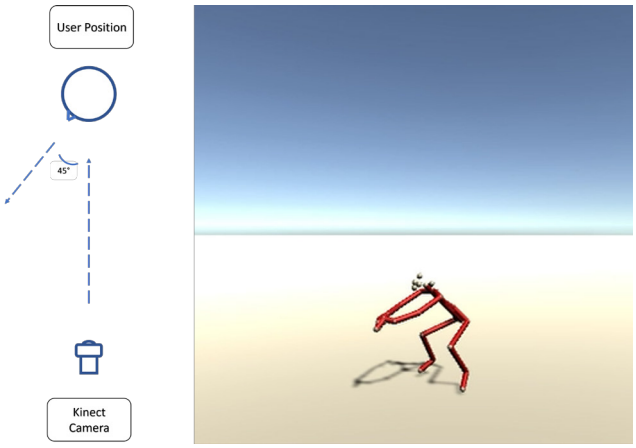


Figure 2: User position during application evaluation. *(Left)* A top-down view of the camera and user position. *(Right)* The view of the user from the camera's perspective.

2.3.3. Procedure

The camera was placed on a stand approximately four feet in height and connected to a laptop PC via USB-C. Each participant was asked to wear a pair of Bose wireless Bluetooth headphones and listen to each sound mapping while they performed a squat. The users were allowed to complete as many squats as needed in order to understand how the sound was mapped to the movement. Once the user felt that they had a good idea of the mapping, they moved onto the next sonic design and repeated the process. After they explored all five mappings, each participant was asked to complete a brief questionnaire, in which they selected their preferred mapping based on its intuitiveness and aesthetic. The questions within the questionnaire were as follows:

- Q1: Which feedback did you prefer the most?
- Q2: Which feedback did you feel was most informative? (i.e. which gave the most information about how your body was moving in space?)
- Q3: Which feedback would you be able to listen to for an extended period of time (20min)?

2.3.4. Results

Table 1 presents a summary of the responses to the questionnaire and shows that the wooden keyboard was the most aesthetically pleasing sonic mapping and the sine wave was the most informative. Additionally, some participants offered verbal feedback. Two participants stated that even though they enjoyed listening to both the wooden keyboard and whistling bottles, they reported feeling unsure of what the feedback was instructing them to do. One participant mentioned that in addition to the conch trumpet's pleasant sound, he specifically enjoyed that the sound "turned off and gave his ears a break" as he returned to the standing position.

Participant No.	Q1: Preference	Q2: Informative	Q3: Extended Listening
1	Whistling Bottles	Sine Wave	Whistling Bottles
2	Conch Trumpet	Conch Trumpet	Conch Trumpet
3	Wooden Keyboard	Sine Wave	Wooden Keyboard
4	Wooden Keyboard	Conch Trumpet	Wooden Keyboard

Table 1: Participant responses to sonification questionnaire.

2.3.5. Discussion

This evaluation provided valuable insight into what sonic mappings were most preferable and informative during application use. Although the whistling bottles and the wooden keyboard were reported as pleasant to listen to, their mappings were not readily understood. The sine wave was reported as informative, but not pleasant to listen to. Additionally, the ability of high frequency, or high pitch, sounds to induce stress has been corroborated in previous research on perceptual sonification [27, 28]. Having a higher pitch present while the user is in the squatted position could result in them feeling an increased sense of urgency return to their standing position, resulting in uneven motion. We want our users to develop a steady and even squat movement, therefore, a more pleasant sonification would be better accommodating of even movement and pacing. We chose to move forward with the conch trumpet because it was reported as both informative and pleasant by the participants. The conch trumpet was also one of our contextual feedback mappings. Its sound intensified as the user approached their target knee angle flexion and dissipated as they returned to their standing position. This contextual feedback approach has been shown to be an optimal paradigm for learning. [25]. It should also be noted that the sonic designs were not randomized when administered. We understand that due to the lack of counterbalancing there is a possibility that our results could suffer from order effects.

3. ITERATION 2

3.1. Design Modification

We expanded the set of sonified movement features to incorporate foot placement, knee alignment and weight shifting.

Keeping the technical setup identical to the first iteration, automated verbal instructions and sonification feedback provided the user with guidance as they fulfilled all four squat requirements: foot placement, knee flexion, knee alignment, weight shifting. Real-time feedback was provided concurrently with the motion, to assist users in fine-tuning the aforementioned movement features when performing the squat.

On startup, the user was first asked to complete a brief calibration, during which the user's target knee flexion angle range was determined. This calibration was intended to be completed in the presence of a physical therapist or trainer as it records the values

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needed to make sure the user was performing within their appropriate range of motion and actively engaging and challenging their body. The maximum angle was the angle between the user's thigh and shank while the knee was fully extended and the minimum angle was the angle between the user's thigh and shank while the knee was flexed at the user's target flexion angle. These values were dependent on each user's range of motion and ranged between 180 (extension) to 90 (flexion) degrees. After, this one-time calibration, no further input was required from a physiotherapist or trainer.

Next, the user was asked to place their feet shoulder width apart. Once this was achieved, they heard a bell signifying that the correct position had been reached. The users then received a verbal instruction to maintain this foot placement and lower themselves into the squatting position as if sitting back on a chair.

The training mode of the application was split into three repetitions, which grew in complexity with each repetition, in order to have the user focus on different aspects of the movement (refer Fig. 3). We believed a layered approach would help users better understand what information was being provided to them and avoid overly dense feedback at the beginning of the training. During the second repetition, the corresponding sonification feedback was used to drive user attention towards ensuring their knees were not moving past their toes as they performed the squat. During the third repetition, the corresponding sonification feedback was used to drive user attention towards how their weight was shifting throughout the exercise. Once the three repetitions were completed, the user had the option to repeat the exercises with verbal instruction or select an alternative mode that allowed them to train with only the sonification feedback and no verbal instruction.

3.1.1. Technical Development

The technical development and setup remained the same as in Section 2.1.1.

3.2. Implementation Modification

3.2.1. Movement Feature Computation

Foot Placement: To determine if the user's feet are shoulder width apart, we used the left and right foot positions to calculate the distance between the feet. We also used the left and right shoulder positions to calculate the distance between the shoulders. We then compared the calculated shoulder distance to the foot distance to determine if the feet were in the correct position.

Knee Alignment: The knee alignment was monitored using the left and right knee and toe positions. We compared the values of the x-position of the knee with the x-position of the toe to determine if the knee extended forward beyond the toe and outside of recommended knee alignment.

Weight Shifting: Weight shifting was assessed via the movement of the user's *center of mass (CoM)*. The CoM position was calculated using the segmentation method found in Lui et. al.'s [26] paper on novel lower-limb rehabilitation [26]. The method used the mass and segment midpoints of different body segments to calculate the CoM for each limb, independently. The CoM of the entire body was then calculated from these individual CoMs. We generalized the values necessary for these calculations by using anthropometric data [26].

3.2.2. Sonic Interaction Design and Mapping

Based on our previous findings that users preferred the conch trumpet, we chose a similar physical model with more parametric control - the flute. The flute instrument model provided parameters corresponding to mouth position and tube length of the instrument, in addition to the blowing pressure excitation applied to the model.

Foot Placement: The distance between the user's feet was linearly mapped to the tube length of the flute, which ranged between, 0.01 to 3 meters. The pitch was changed by modulating the length of the tube of the instrument. As the distance between the feet increased, the tube length also increased, resulting in a decrease in pitch. As the distance between the feet decreased, the tube length decreased, resulting in an increase in pitch. For this mapping, we again relied on the logic regarding pitch and urgency explored for perceptual sonification [27, 28]. The sense of urgency invoked during the this stage would ideally motivate the user towards finding the correct foot placement in a quick and efficient manner.

Knee Flexion: The knee flexion angle was linearly mapped to the blowing pressure parameter of the physical model. As the user approached their target knee flexion angle, the blowing pressure increased. The blowing pressure decreased to zero, as the user returned to full knee extension. This mapping was intended to give the user a sense of control over the virtual instrument by correlating the depth of their squat with the intensity of sound produced.

Knee Alignment: The user's knee alignment was represented by the presence and absence of white noise feedback. The presence of noise is effective in stopping excess movement beyond an undesirable target point [22]. Therefore, if the user's knees moved beyond their toes, the salient and immediate noise feedback was intended to signify to the user that any excess movement of the knees beyond the toes needed to be rectified immediately.

Weight Shifting: The user's CoM was linearly mapped to the mouth position on the flute physical model. If the COM shifted beyond the user's base of support (calculated here as the range between their heels and feet), the mouth position on the physical model would change causing an overblowing effect that resulted in an undesirable pitch shift. This shift causes the flute to sound extremely shrill. This mapping was intended to correlate instances of uneven weight shifting physically with the shifting of the pitch to a shrill octave.

Videos samples of the mappings can be found here[6]. In addition to the new flute physical model, we also incorporated the triggering of sound samples [7], one to notify the users that the target knee flexion has been reached, and the other to signify the return to the starting position was reached, signifying the completion of the repetition.

3.3. Evaluation 2: Usability Evaluation

3.3.1. Participants

Four male participants provided feedback on the usability of the application. All of the participants had a background in sound and music computing. All participants were volunteers and consented to being included in the evaluation. No sensitive information was collected.

[6] https://drive.google.com/drive/folders/1EbruQ7OeN_KB6Wfsq9FHaVZT_kB-Kxj9?usp=sharing

[7] https://drive.google.com/drive/folders/1EW7GKVXYKrOd1v9XQOZSB247znwTgMIH?usp=sharing

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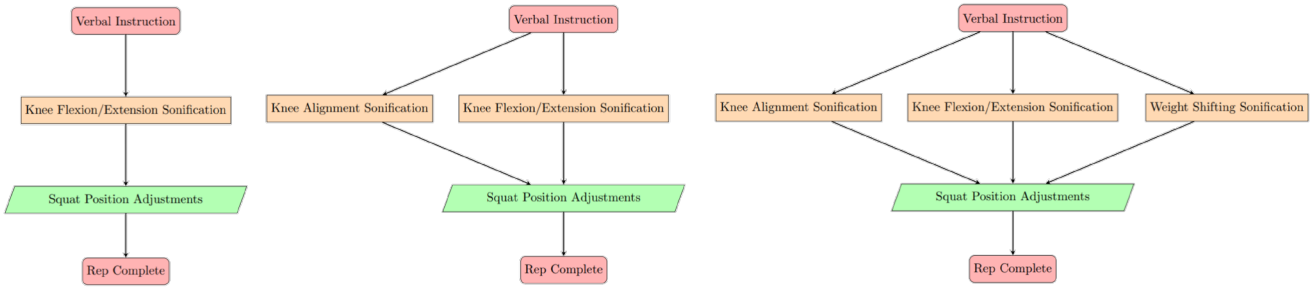


Figure 3: Flowchart depictions of sonification additions during repetitions after the target foot distance is achieved. (*Left*) Repetition 1: Squat flexion/extension sonification is activated, (*Middle*) Repetition 2: Knee alignment sonification is activated, and (*Right*) Repetition 3: Weight shifting sonification is activated.

3.3.2. Experimental Setup

The evaluation was carried out in a naturally lit room at the 2022 Sound and Music Computing Conference. Participants were requested to stand on a marker six feet away, with their body rotated as previously described in section 2.3.2.

3.3.3. Procedure

The camera was placed on a desk approximately three feet in height and connected to a laptop PC via USB-C. Each user completed an individual calibration, administered by the researchers, after which they were asked to put on a pair of Bose wireless Bluetooth headphones and complete the designed interaction developed for our prototype. After they completed the interaction, each participant completed the System Usability Scale Questionnaire(SUS) [29]. There were two additional statements added to the SUS questionnaire, regarding the intuitiveness and aesthetic values of the sonic feedback.

- S1: The feedback provided by the system was informative and helpful.
- S2: I would be able to listen to this feedback for an extended period of time.

The responses for these two statements were recorded on a Likert scale, where 1 was strongly disagree and 5 was strongly agree.

3.3.4. Results

The application was calculated to have a mean SUS score of 78.75, equating to a grade of B, ”good” usability. The application received a rating of 4 or higher for each positive SUS statement except SUS1: I think that I would like to use this system frequently and SUS9: I felt very confident using the system. The application received a minimum and maximum score for 62.5 and 95, respectively, with a standard deviation of 14.2. Table 2 presents a summary of the responses to the individual items of the SUS questionnaire.

One participant provided additional verbal feedback. The participant expressed interest in having a verbal reminder if the target position was not met within a specific time frame, while interacting with the application.

3.3.5. Discussion

The results of the usability evaluation indicate that the participants largely agreed that the sonification was informative and listenable. Although the SUS score and sonic design ratings were above average, we do understand that this score could be subject to social desirability bias as the data were not collected in a blinded manner. Even though the ratings for the statements were moderately above average, the feedback received for SUS1 and SUS9 immediately identifies an area for improvement on our application. We want to design a system that the user will be eager to use, therefore future iterations of this work should focus on incorporating gamification techniques, as they have been shown to increase motivation and engagement [30, 31].

4. GENERAL DISCUSSION

In this paper, we presented the development of an auditory-only application that emphasizes real-time movement sonification feedback for bodyweight squat learning and performance that can be operated without the need for in-person training. Our work expands on previous studies through the utilization of a real-time concurrent sonification scheme that provides feedback information for multiple movement parameters. Additionally, we provided the sonic feedback in a logical, sequential order. To our knowledge, this is the first study to develop an application incorporating these different features. Our concurrent, sequential sonification scheme provides users with instantaneous task-relevant feedback to ensure they are performing the squat movement safely and correctly[14]. The use of visual feedback, in other studies diverts attention from the movement form, as the user may need to process information presented on visual displays[12]. Our use of concurrent sonification, allows the user to focus their attention on the necessary position changes or alterations critical for proper squat form [14]. Ideally, our application will assist users in developing a greater physical connection and understanding of the squat exercise and through repeated training prepare them to more easily identify any errors they may have in their squat form.

We conducted two evaluations during this study. The first was an evaluation of the sonic design mappings. The results from our evaluations confirmed our expectations that contextual feedback mappings would be more suitable for squat training. The second evaluation assessed the application’s usability. Our mean SUS

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Participant No.	SUS1	SUS2	SUS3	SUS4	SUS5	SUS6	SUS7	SUS8	SUS9	SUS10	SUS, Grade
1	4	1	5	3	4	2	5	1	5	2	85, A
2	3	1	3	4	5	1	3	2	2	3	62.5, D
3	4	2	4	1	4	2	3	2	3	2	72.5, B
4	4	1	5	1	5	1	5	1	4	1	95, A
Mean Ratings	3.75	1.25	4.25	2.25	4.5	1.5	4	1.5	3.5	2	78.75, B

Table 2: Individual participant ratings for SUS questionnaire. The SUS questionnaire included 10 items. The SUS alternates between odd-numbered positive statements, for which a high score is desirable, and even numbered negative statements, for which a low score is desirable, rated on a Likert scale from 1 (strongly disagree) to 5 (strongly agree). SUS 1: I think that I would like to use this system frequently; SUS2: I found the system unnecessarily complex; SUS3: I thought the system was easy to use; SUS4: I think that I would need the support of a technical person to be able to use this system; SUS5: I found the various functions in this system were well integrated; SUS6: I thought there was too much inconsistency in this system; SUS7: I would imagine that most people would learn to use this system very quickly; SUS8: I found the system very cumbersome to use; SUS9: I felt very confident using the system; SUS10: I needed to learn a lot of things before I could get going with this system. The color coding corresponds to the valence of the statements in the questionnaire: positive (green) and negative (red).

score of 78.75 indicates good usability and reinforces the use of this sonification paradigm as a new approach for multi-targeted movement sonification. Although our results show promised for our application development, there were several limitations identified in our work.

The background of the participants was a limitation of these evaluations. All of the participants had some familiarity with sound engineering that may have made them more apt at understanding what parameters were modulated as the sounds were played. Since there were no users who did not have this experience, we are unsure how readily users without this background would have been able to hear and understand the sound modulations as they squatted. Additionally, all of the participants were male and their ages were not provided, therefore, we were unable to gain insight to usability preferences within a female population or perform age and usability correlations. We designed this application to be accessible and informative for a wider range of users. Ideally, further iterations of this work will include women, recreational athletes, the elderly, and individuals with visual impairments in the participant pool, in efforts to accrue more varied feedback on the usability of the application.

Future work should conduct a more robust randomized control study to assess short-term performance improvement and long-term retention rates when training with the application. Additionally, future work should test the sonic designs not only for aesthetics and informativeness, but also for instructional value by having participants complete a target finding exercise [32]. This would allow researchers to assess the effectiveness of their designs in multidimensional planes. Lastly, we would also like to expand this workflow to additional exercises, in order to create a full, thorough workout regimen.

5. CONCLUSIONS

The goal of this study was to develop a real-time movement sonification paradigm for squat training. This application, once calibrated, is simple to use and reduces the need for in person training, allowing users to exercise safely and at their own discretion. We designed an application and implemented concurrent, sequential audio feedback for the users. The application was evaluated and we uncovered several merits of our sonification scheme. This project is one of the first of its kind and our results imply that with future improvements this application has the potential to assist users in quickly identifying misalignment or impingement of joints and muscles [3, 4], strengthening several lower body muscle groups at once [1, 6] and aid a wide variety of users in their pursuit of better health and exercise. The Squat Training repository is available here[8].

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